Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Course: B.Tech
Code: 105302

Subject: Data Structure and Algorithms

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) What is the time complexity of deleting a linked list?
- (i) O (1)
- (ii) O (n)
- (iii) O (log n)
- (iv) O (n2)

(b) Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are
- (i) 4 and 15 respectively
- (ii) 3 and 14 respectively
- (iii) 4 and 14 respectively
- (iv) 3 and 15 respectively

(c) Which sorting algorithm is stable and works efficiently for small or nearly sorted datasets?
- (i) Quick Sort
- (ii) Heap Sort
- (iii) Insertion Sort
- (iv) Selection Sort

(d) In a BST, deletion of a node with two children uses
- (i) Random node
- (ii) Inorder successor
- (iii) Any one of leaf nodes
- (iv) Root node

(e) In a doubly linked list, what is the minimum pointer updates required to delete a node (not head/tail)
- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

(f) Which of the following pairs of traversals is not sufficient to build a binary tree from the given traversals?
- (i) Preorder and Inorder
- (ii) Inorder and Postorder
- (iii) Preorder and Postorder
- (iv) None of the above

(g) The number of edges in a complete graph with 'n' vertices is?
- (i) $n(n-1)$
- (ii) n^2
- (iii) $n(n-1)/2$
- (iv) $2n-1$

(h) Consider a B+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?
- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

(i) Given a hash table T with 25 slots that stores 2000 elements, the load factor α for T is
- (i) 80
- (ii) 0.0125
- (iii) 8000
- (iv) 1.25

(j) Binary search
- (i) is divide and conquer method
- (ii) can be implemented using recursive function
- (iii) is better than linear search in worst case
- (iv) all of these

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Q.2 (a) Explain the role of time–space trade-off while choosing suitable data structure for a real-world application. Take suitable examples. [7]

(b) How Abstract Data Types achieve abstraction? Illustrate your answer with the help of Queue ADT, explaining both physical and abstracted views. [7]

Q.3 (a) Convert the following infix expression to equivalent postfix expression using stack. Also, evaluate the postfix expression using stack: $7 + 5 * 3^2/(9 - 2^2) + 6 * 4$. Clearly mention the action taken and stack-state for each symbol scanned from input expression. [7]

(b) What is Circular queue? Write C functions for Q-insert and Q-delete operations on a circular queue. [7]

Q.4 (a) What is Binary Search Tree? Write a C function to insert a given node into BST. [7]

(b) Compare array-based lists and linked lists. Design an algorithm to reverse a singly linked list. [7]

Q.5 (a) Write a C program to implement stack operations using linked list. Analyse the implementation complexity also. [7]

(b) Write an algorithm to insert an element somewhere in the middle of a doubly linked list after given nth node. [7]

Q.6 (a) If the following sequence of numbers is to be sorted using quick sort, then show the iterations/passes of the sorting process: $42, 34, 75, 23, 21, 18, 90, 67, 78$. [7]

(b) Write an algorithm for merge sort technique. Derive the best, average and worst complexity of your algorithm also. [7]

Q.7 (a) Explain the concept of hashing and its importance in data structures. Discuss different collision resolution techniques used in hashing. [7]

(b) How does an AVL-tree differ from binary search tree? Insert the following keys in the order given below to build an AVL-tree: [7]

$$12, 11, 13, 10, 9, 15, 14, 18, 7, 6, 5, 4$$

Clearly mention different rotations used showing balance factor of each node.

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Q.8 (a) The level order traversal of a max heap is $50, 40, 30, 25, 16, 23, 20$. What will be the level order traversal after inserting the elements 28, 43, 11 sequentially. Clearly indicate the heapify steps after each insertion. [7]

(b) Discuss various tree traversal schemes with suitable example. Write a recursive function for in-order tree traversal. [7]

Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Asymptotic Notations
(b) BFS and DFS
(c) B-Tree
(d) Priority Queue

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